

GIT-HOL-3:-

HandsOn:-

- Prerequisites:

```
C:\Users\riyak>cd GitDemo

C:\Users\riyak\GitDemo>git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        welcome.txt

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\riyak\GitDemo>git add .

C:\Users\riyak\GitDemo>git commit -m "Initail Commit"
[main (root-commit) 1f514e4] Initail Commit
 2 files changed, 3 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 welcome.txt

C:\Users\riyak\GitDemo>
```

- Part 1: Branching
 1. Step 1: Create a new branch

```
C:\Users\riyak\GitDemo>git branch GitNewBranch

C:\Users\riyak\GitDemo>git branch
GitNewBranch
* main

C:\Users\riyak\GitDemo>
```

2. List all local and remote branches

```
C:\Users\riyak\GitDemo>git branch -a
GitNewBranch
* main

C:\Users\riyak\GitDemo>
```

3. Switch to the new branch

```
C:\Users\riyak\GitDemo>git checkout GitNewBranch
Switched to branch 'GitNewBranch'

C:\Users\riyak\GitDemo>git branch
* GitNewBranch
  main
```

4. Create a file

```
C:\Users\riyak\GitDemo>echo This file belongs to GitNewBranch > branch.txt

C:\Users\riyak\GitDemo>dir
Volume in drive C is OS
Volume Serial Number is 00FD-534E

Directory of C:\Users\riyak\GitDemo

22-07-2026  22:40    <DIR>          .
22-07-2026  22:39    <DIR>          ..
22-07-2026  22:09                11 .gitignore
22-07-2026  22:07                24 application.log
22-07-2026  22:40                36 branch.txt
22-07-2026  22:07    <DIR>          log
22-07-2026  21:49                33 welcome.txt
                4 File(s)              104 bytes
                3 Dir(s)  8,004,653,056 bytes free
```

5. Check Git Status

```
C:\Users\riyak\GitDemo>git status
On branch GitNewBranch
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    branch.txt

nothing added to commit but untracked files present (use "git add" to track)
```

6. Add the File

```
C:\Users\riyak\GitDemo>git add branch.txt
```

7. Commit

```
C:\Users\riyak\GitDemo>git commit -m "Added branch.txt in GitNewBranch"
[GitNewBranch db5b951] Added branch.txt in GitNewBranch
 1 file changed, 1 insertion(+)
 create mode 100644 branch.txt
```

8. Verify

```
C:\Users\riyak\GitDemo>git add branch.txt
C:\Users\riyak\GitDemo>git commit -m "Added branch.txt in GitNewBranch"
[GitNewBranch db5b951] Added branch.txt in GitNewBranch
 1 file changed, 1 insertion(+)
 create mode 100644 branch.txt
C:\Users\riyak\GitDemo>git status
On branch GitNewBranch
nothing to commit, working tree clean
```

- Part 2: Merging

1. Switch back to main

```
C:\Users\riyak\GitDemo>git status
On branch GitNewBranch
nothing to commit, working tree clean

C:\Users\riyak\GitDemo>git checkout main
Switched to branch 'main'

C:\Users\riyak\GitDemo>git branch
  GitNewBranch
* main
```

2. Show differences(Command Line)

```
C:\Users\riyak\GitDemo>git diff main GitNewBranch
diff --git a/branch.txt b/branch.txt
new file mode 100644
index 0000000..18133c5
--- /dev/null
+++ b/branch.txt
@@ -0,0 +1 @@
+This file belongs to GitNewBranch
```

3. Show visual differences

4. Merge the branch

```
PS C:\Users\riyak\GitDemo> git merge GitNewBranch
Updating 1f514e4..db5b951
Fast-forward
 branch.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 branch.txt
PS C:\Users\riyak\GitDemo> |
```

5. Verify merge

```
PS C:\Users\riyak\GitDemo> dir

Directory: C:\Users\riyak\GitDemo

Mode                LastWriteTime         Length Name
----                -
d-----         22-07-2026        22:07          log
-a-----         22-07-2026        22:09          11 .gitignore
-a-----         22-07-2026        22:07          24 application.log
-a-----         22-07-2026        22:53          36 branch.txt
-a-----         22-07-2026        21:49          33 welcome.txt

PS C:\Users\riyak\GitDemo> |
```

6. View commit history

```
PS C:\Users\riyak\GitDemo> git log --oneline --graph --decorate
* db5b951 (HEAD -> main, GitNewBranch) Added branch.txt in GitNewBranch
* 1f514e4 Initail Commit
PS C:\Users\riyak\GitDemo> |
```

7. Delete the merge branch

```
PS C:\Users\riyak\GitDemo> git branch -d GitNewBranch
Deleted branch GitNewBranch (was db5b951).
PS C:\Users\riyak\GitDemo> |
```

8. Verify

```
PS C:\Users\riyak\GitDemo> git branch
* main
PS C:\Users\riyak\GitDemo> |
```

9. Check Status

```
PS C:\Users\riyak\GitDemo> git status
On branch main
nothing to commit, working tree clean
PS C:\Users\riyak\GitDemo> |
```